

-- PART 1: DDL

CREATE DATABASE company_db;

SHOW DATABASES;

USE company_db;

```
CREATE TABLE employees (  
    emp_id INT,  
    employee_name VARCHAR(30),  
    department VARCHAR(30),  
    salary INT  
);
```

DESC employees;

ALTER TABLE employees ADD email VARCHAR(50);

ALTER TABLE employees ADD joining_date DATE;

DESC employees;

ALTER TABLE employees MODIFY employee_name VARCHAR(50);

ALTER TABLE employees RENAME COLUMN employee_name TO emp_name;

ALTER TABLE employees ADD PRIMARY KEY (emp_id);

ALTER TABLE employees ADD UNIQUE (email);

ALTER TABLE employees ADD CHECK (salary > 10000);

ALTER TABLE employees RENAME TO employee_details;

CREATE TABLE departments ();

ALTER TABLE departments ADD dept_id INT;

ALTER TABLE departments ADD dept_name VARCHAR(30);

SHOW TABLES;

DROP TABLE departments;

SHOW TABLES;

```
DROP TABLE employee_details;
```

```
-- PART 2: DML
```

```
CREATE TABLE employee_details (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    department VARCHAR(30),  
    salary INT CHECK (salary > 10000),  
    email VARCHAR(50) UNIQUE,  
    joining_date DATE  
);
```

```
INSERT INTO employee_details VALUES  
(101, 'Sai', 'IT', 50000, 'sai@gmail.com', '2024-01-10');
```

```
INSERT INTO employee_details VALUES  
(102, 'Anjali', 'HR', 42000, 'anjali@gmail.com', '2024-02-12');
```

```
INSERT INTO employee_details VALUES  
(103, 'Ravi', 'Finance', 55000, 'ravi@gmail.com', '2024-03-15'),  
(104, 'Sneha', 'Marketing', 48000, 'sneha@gmail.com', '2024-04-18'),  
(105, 'Karthik', 'Sales', 46000, 'karthik@gmail.com', '2024-05-20');
```

```
INSERT INTO employee_details VALUES  
(106, 'Suresh', 'IT', 60000, 'suresh@gmail.com', '2024-06-01');
```

```
INSERT INTO employee_details VALUES  
(107, 'Harika', 'HR', 40000, 'harika@gmail.com', '2024-06-05');
```

```
INSERT INTO employee_details VALUES  
(108, 'Manoj', 'Finance', 52000, 'manoj@gmail.com', '2024-06-10');
```

```
INSERT INTO employee_details VALUES  
(109, 'Varun', 'Marketing', 47000, 'varun@gmail.com', '2024-06-15');
```

```
INSERT INTO employee_details VALUES  
(110, 'Arjun', 'Sales', 43000, 'arjun@gmail.com', '2024-06-20');
```

```
UPDATE employee_details  
SET salary = 58000  
WHERE emp_id = 101;
```

```
UPDATE employee_details  
SET department = 'IT'
```

```
WHERE emp_id = 105;
```

```
UPDATE employee_details  
SET email = 'ravi.new@gmail.com'  
WHERE emp_id = 103;
```

```
UPDATE employee_details  
SET salary = salary + 5000  
WHERE department = 'IT';
```

```
UPDATE employee_details  
SET salary = salary + (salary * 0.10)  
WHERE department = 'HR';
```

```
UPDATE employee_details  
SET department = 'Marketing'  
WHERE department = 'Sales';
```

```
DELETE FROM employee_details  
WHERE emp_id = 110;
```

```
DELETE FROM employee_details  
WHERE department = 'Finance';
```

```
INSERT INTO employee_details VALUES  
(111, 'Deepak', 'Finance', 56000, 'deepak@gmail.com', '2024-07-01'),  
(112, 'Priya', 'Finance', 59000, 'priya@gmail.com', '2024-07-05');
```

```
DELETE FROM employee_details  
WHERE salary < 45000;
```

```
UPDATE employee_details  
SET salary = salary + 3000  
WHERE department = 'Marketing';
```

```
DELETE FROM employee_details;
```

```
-- PART 3: DQL  
-- Insert fresh data first
```

```
INSERT INTO employee_details VALUES  
(101, 'Sai', 'IT', 55000, 'sai@gmail.com', '2024-01-10'),  
(102, 'Anjali', 'HR', 45000, 'anjali@gmail.com', '2024-02-12'),  
(103, 'Ravi', 'Finance', 62000, 'ravi@gmail.com', '2024-03-15'),  
(104, 'Sneha', 'Marketing', 48000, 'sneha@gmail.com', '2024-04-18'),  
(105, 'Karthik', 'IT', 70000, 'karthik@gmail.com', '2024-05-20'),
```

```
(106, 'Suresh', 'IT', 60000, 'suresh@gmail.com', '2024-06-01'),  
(107, 'Harika', 'HR', 52000, 'harika@gmail.com', '2024-06-05'),  
(108, 'Manoj', 'Finance', 58000, 'manoj@gmail.com', '2024-06-10'),  
(109, 'Varun', 'Marketing', 50000, 'varun@gmail.com', '2024-06-15'),  
(110, 'Sanjana', 'HR', 47000, 'sanjana@gmail.com', '2024-06-20');
```

```
SELECT * FROM employee_details;
```

```
SELECT emp_name FROM employee_details;
```

```
SELECT emp_name, salary FROM employee_details;
```

```
SELECT * FROM employee_details  
WHERE department = 'IT';
```

```
SELECT * FROM employee_details  
WHERE department = 'HR';
```

```
SELECT * FROM employee_details  
WHERE salary > 50000;
```

```
SELECT * FROM employee_details  
WHERE salary < 60000;
```

```
SELECT * FROM employee_details  
WHERE salary BETWEEN 45000 AND 60000;
```

```
SELECT * FROM employee_details  
WHERE department = 'IT' AND salary > 50000;
```

```
SELECT * FROM employee_details  
WHERE department = 'HR' OR department = 'Finance';
```

```
SELECT * FROM employee_details  
WHERE department != 'HR';
```

```
SELECT * FROM employee_details  
WHERE department IN ('IT', 'Finance');
```

```
SELECT * FROM employee_details  
WHERE emp_name LIKE 'S%';
```

```
SELECT * FROM employee_details  
WHERE emp_name LIKE '%a';
```

```
SELECT * FROM employee_details  
WHERE emp_name LIKE '%ar%';
```

```
SELECT DISTINCT department FROM employee_details;
```

```
SELECT * FROM employee_details  
ORDER BY salary ASC;
```

```
SELECT * FROM employee_details  
ORDER BY salary DESC;
```

```
SELECT * FROM employee_details  
ORDER BY emp_name;
```

```
SELECT * FROM employee_details  
LIMIT 5;
```

-- BONUS DQL: Aggregate Functions

```
SELECT COUNT(*) AS total_employees  
FROM employee_details;
```

```
SELECT SUM(salary) AS total_salary  
FROM employee_details;
```

```
SELECT AVG(salary) AS average_salary  
FROM employee_details;
```

```
SELECT MAX(salary) AS highest_salary  
FROM employee_details;
```

```
SELECT MIN(salary) AS lowest_salary  
FROM employee_details;
```

```
SELECT department, COUNT(*) AS employee_count  
FROM employee_details  
GROUP BY department;
```

```
SELECT department, SUM(salary) AS total_salary  
FROM employee_details  
GROUP BY department;
```

```
SELECT department, AVG(salary) AS average_salary  
FROM employee_details  
GROUP BY department;
```

```
SELECT department, COUNT(*) AS employee_count
```

```
FROM employee_details  
GROUP BY department  
HAVING COUNT(*) > 2;
```

```
SELECT department, AVG(salary) AS average_salary  
FROM employee_details  
GROUP BY department  
ORDER BY average_salary DESC  
LIMIT 1;
```